IOT BASED SMART WALKING CANE FOR OBSTACLE DETECTION AND SAFETY ENHANCEMENT

Cohort 5

System Architecture

IoT Based Smart Walking Cane for Obstacle Detection and Safety Enhancement

1. **Sensing layer**
   Ultrasonic · IR · Water · Accelerometer
2. **Processing layer**
   ESP32 — sensor fusion · alert logic
3. **Alert output layer**
   Vibration · Buzzer · Bluetooth audio
4. **Communication layer**
   GPS · GSM · WiFi IoT cloud
5. **Power layer**
   LiPo battery · Charger · Boost converter

Sensors
5 modules

Alerts
3 channels

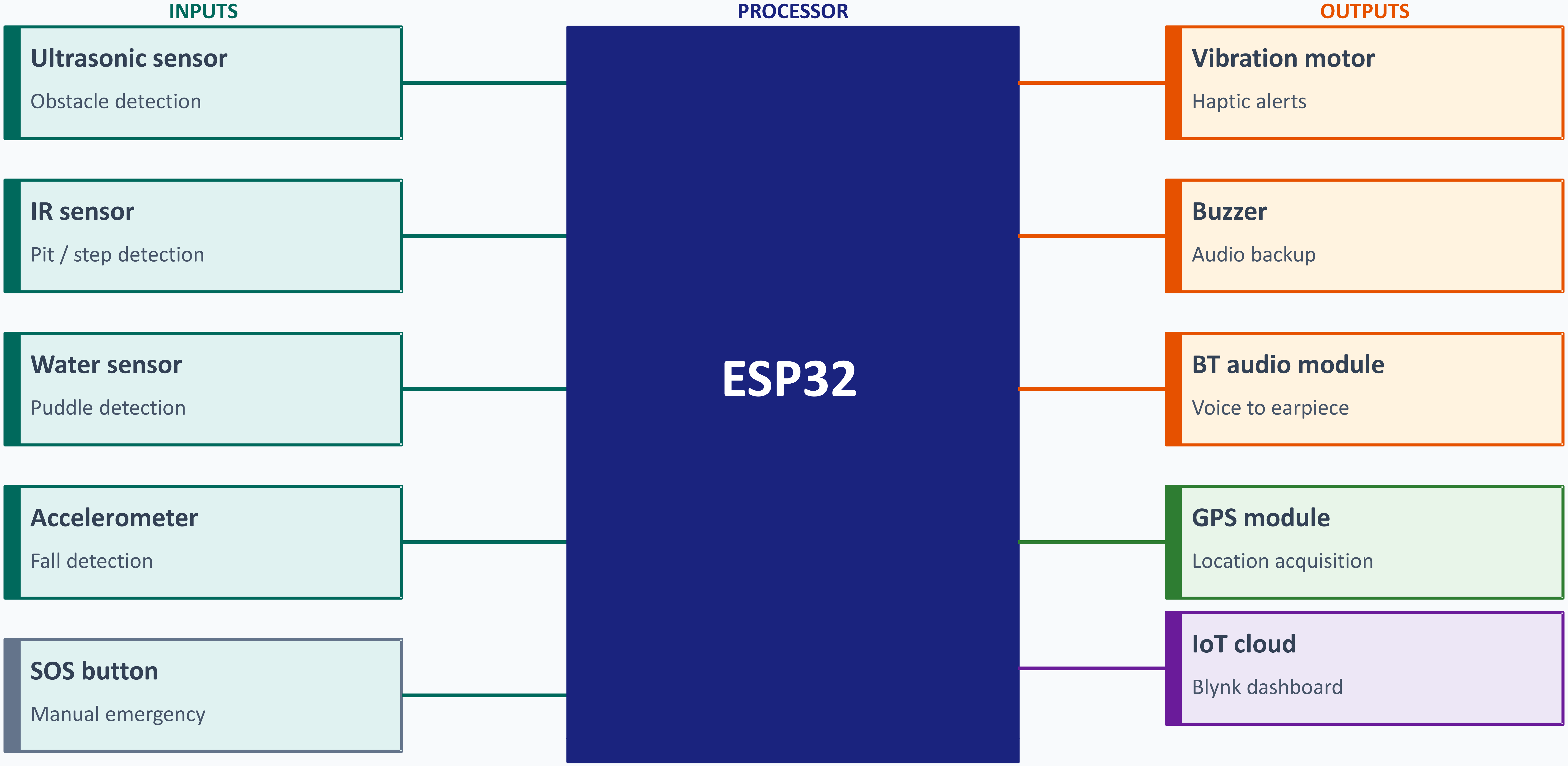
ESP32
Central microcontroller

GPS + GSM
Emergency SOS

IoT Cloud
Blynk / WiFi

Block Diagram

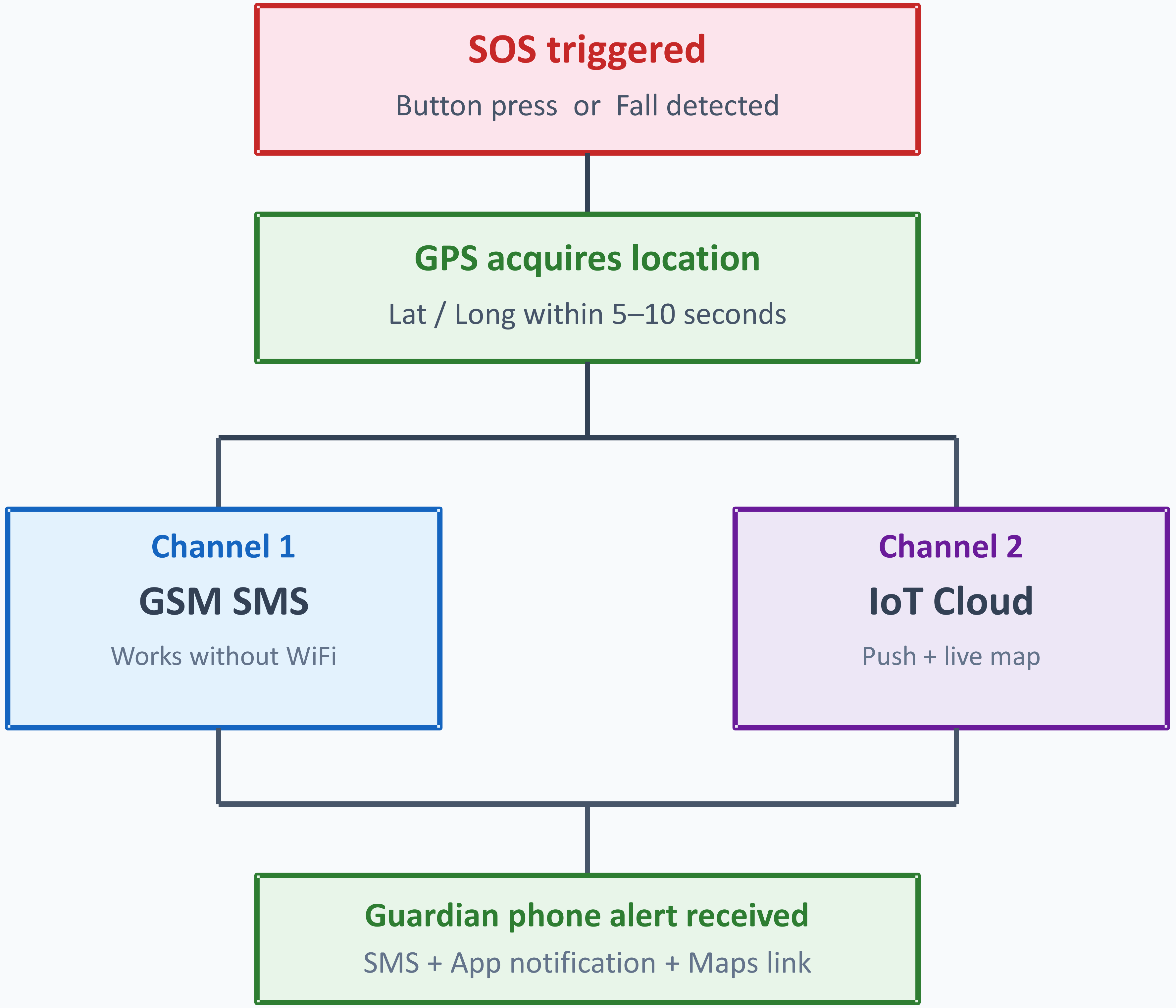
Signal flow from sensors → ESP32 → alerts and communication modules



SOS Flow & Design Decisions

Dual channel emergency alert system and key architectural choices

SOS Alert Flow



Key design decisions

Why ESP32?

Built-in BT A2DP + WiFi eliminates extra modules for audio and IoT

Why waterproof sensors?

Outdoor use in rain requires IP67 rated sensors at cane tip

Why dual SOS channels?

GSM works without WiFi; IoT gives richer data — complementary

Why BT audio?

Private alerts inaudible to bystanders; bone conduction keeps ears open

Why 5-sample filter?

Reduces false alert rate from 15% to under 4% with no hardware cost